Giacomo Comitani

Milan, Italy | comitanigiacomo@gmail.com | github.com/comitanigiacomo
linkedin.com/in/giacomo-comitani | portfolio.jack-lab.dev

Education

University of Milan — Milan, Italy
M.Sc. in Computer Science — *2025 – Present*

- **Main Course Areas**: Artificial Intelligence, Data Analytics & Big Data, Algorithms, Software & Theory, Computing Systems in Industry & Business

University of Milan — Milan, Italy
B.Sc. in Computer Science — *2021 – 2025*

- **Thesis**: *Security Analysis of Obsolete Cryptographic Libraries in Mobile Modems* — Reverse-engineered baseband firmware to evaluate vulnerabilities and vendor patch patterns across mobile devices

Projects

Kanso – Offline-First Habit Tracker | *Go, PostgreSQL, Redis, React, TypeScript, Docker* Jan. 2026 – Mar. 2026

- Engineered a Go sync engine using Hexagonal Architecture, with delta-sync cursors and optimistic concurrency (HTTP 409)
- Implemented Redis rate limiting, async background workers, test suites, and Docker deployment (kanso.jack-lab.dev)

Kafka Fraud Sentinel | *Python, Apache Kafka, mTLS, MongoDB, FastAPI, WebSockets* Aug. 2026 – Sep. 2026

- Built a real-time fraud pipeline on a 3-node Kafka cluster in KRaft mode, evaluating transaction velocity and amount
- Secured end-to-end with mutual TLS (mTLS) via internal CA and ACLs; streamed live alerts via MongoDB and WebSockets

QuickLaunch – GNOME Shell Extension | *JavaScript (GJS), GNOME Shell, Libadwaita* Mar. 2025 – Present

- Created an open-source productivity dock on extensions.gnome.org, engineered with an event-driven model to minimize CPU/battery usage
- Integrated dynamic app search, keyboard shortcuts, and Libadwaita preference windows using JavaScript (GJS)

Personal Homelab & Infrastructure | *Debian, Docker, Nginx, Tailscale, Linux* May 2025 – Present

- Configured a Debian home server hosting containerized services (Nextcloud, Immich, Vaultwarden, Kanso) via Docker Compose
- Hardened infrastructure by isolating all administration and internal traffic strictly behind a private Tailscale mesh network

Experience

ICPC Southwestern Europe Regional Contest (SWERC) — Pisa, Italy
Team Contestant (University of Milan) — *Nov. 2025*

- Represented the university in a 3-person team, solving complex algorithmic and graph problems under strict 5-hour limits

NOI Techpark (SFSCON Hackathon) — Bolzano, Italy
Hackathon Participant — Frontend Developer — *Nov. 2023*

- Collaborated in an intensive 48-hour sprint to build a working open-source prototype, owning the frontend interface

Community Center (Oratorio Dio Trinità d'Amore) — Vimodrone, Italy
Youth Group Leader & Coordinator — *June 2017 – July 2021*

- Organized educational activities and summer camps for youth groups, managing peer coordination, logistics, and group dynamics

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, Java, SQL, Bash, C/C++ (academic), Rust (Beginner)
Backend & Data: PostgreSQL, Redis, Apache Kafka, MongoDB, RESTful APIs, WebSockets, FastAPI
DevOps & Systems: Linux (Debian, Fedora), Docker, Docker Compose, Nginx, Git, GitHub Actions, Tailscale
Concepts & Architecture: Distributed Systems, Hexagonal Architecture, Offline-First Sync, Event-Driven Design, mTLS & ACLs
Spoken Languages: Italian (Native), English (B2)